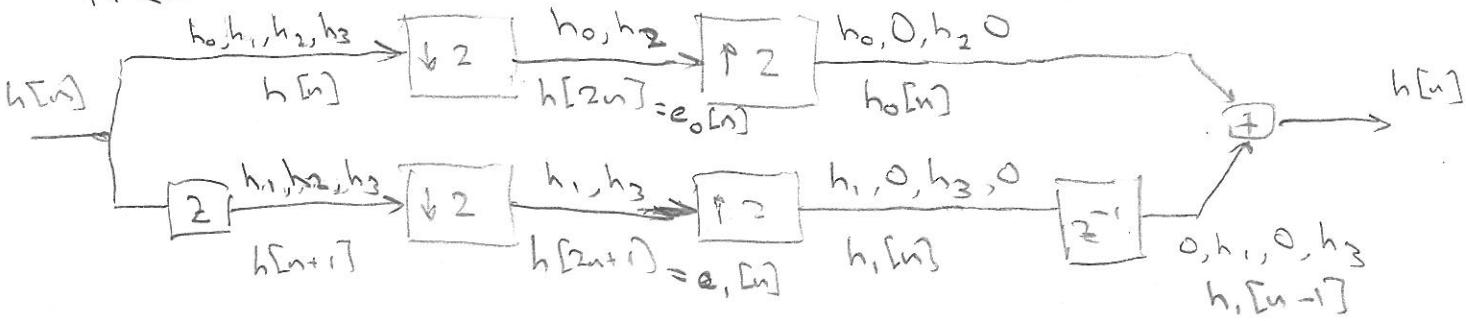
Polyphase Decomposition $h[n] = \{h_0\ h_1\ h_2\ h_3\}$

$M = 2$

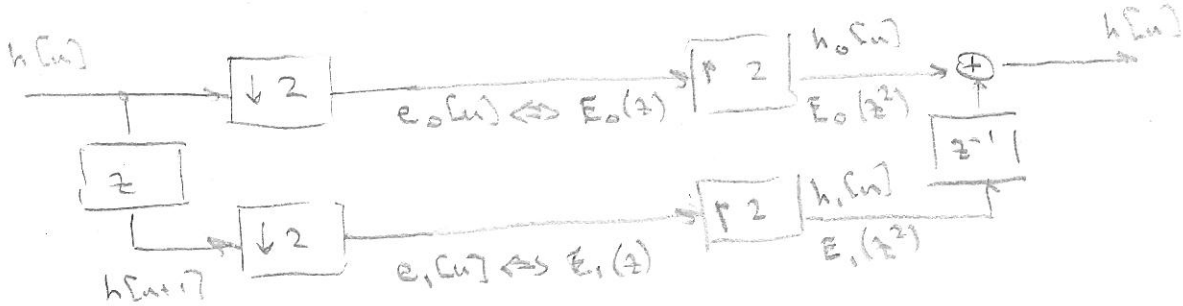


$$h[n] = h_0[n] + h_1[n-1]$$

$$e_0[n] = h[2n] = h_0[2n] \iff E_0(z) = \sum_n e_0[n] z^{-n}$$

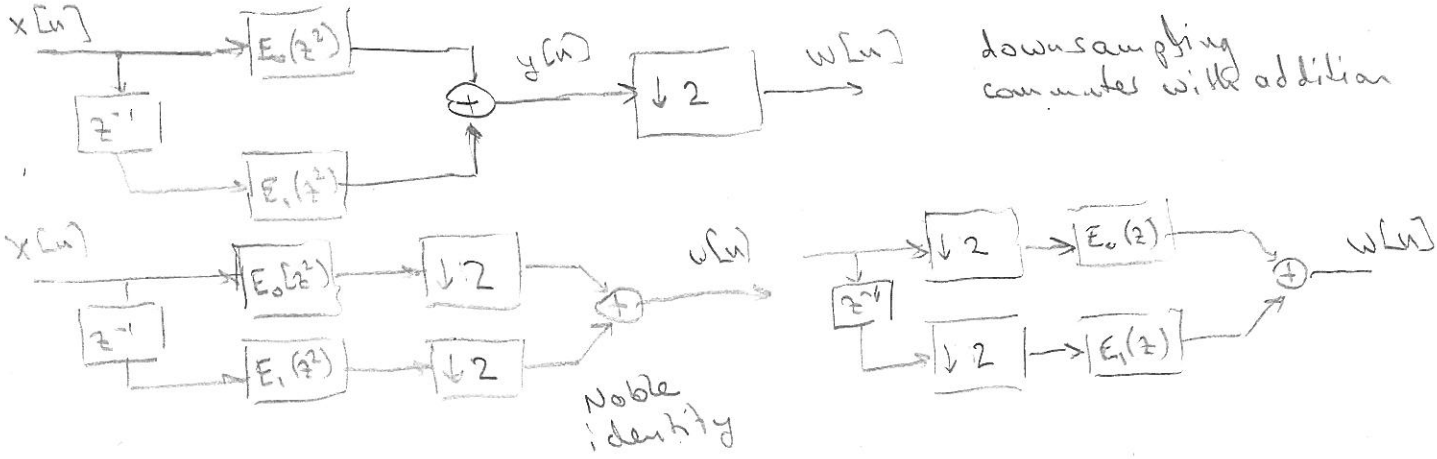
$$e_1[n] = h[2n+1] = h_1[2n] \iff E_1(z) = \sum_n e_1[n] z^{-n}$$

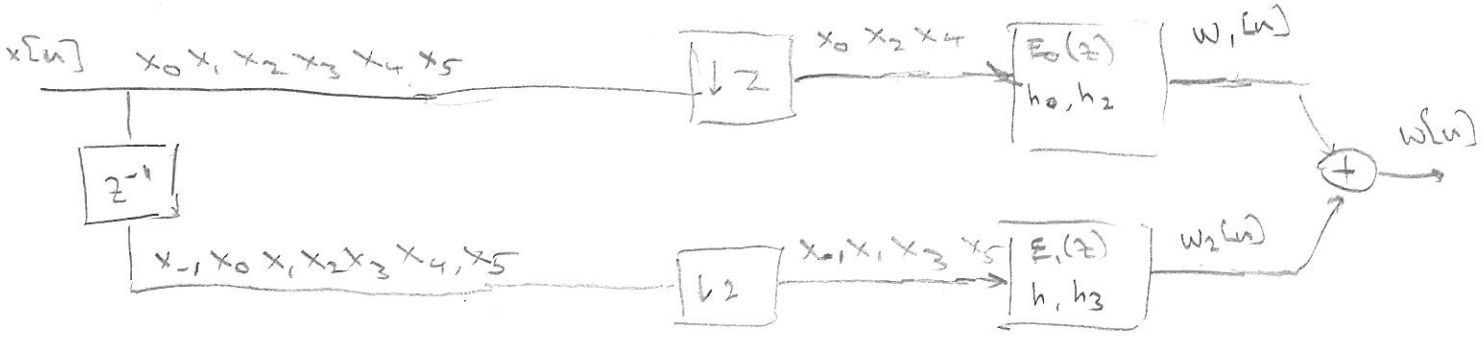
$$h[n] = h_0[n] + h_1[n-1] \iff H(z) = E_0(z^2) + z^{-1} E_1(z^2)$$



$$\boxed{\begin{aligned} H(z) &= E_0(z^2) + z^{-1} E_1(z^2) \\ E_0(z) &= \sum_n h[2n] z^{-n} \\ E_1(z) &= \sum_n h[2n+1] z^{-n} \end{aligned}}$$

$$Y(z) = X(z) \cdot H(z)$$
$$= X(z) \cdot \left[E_0(z^2) + z^{-1} E_1(z^2) \right]$$





$h[n] = \{h_0, h_1, h_2, h_3\}$

$e_0[n] = h[2n]$ $\quad h_0, h_2$

$e_1[n] = h[2n+1]$ $\quad h_1, h_3$

$w_1[n] =$ h_0x_0 h_0x_2 h_0x_4

h_2x_0 h_2x_2 h_2x_4

h_0x_0 $h_0x_2+h_2x_0$ $h_0x_4+h_2x_2$ h_2x_4

$w_2[n] =$ h_1x_{-1} h_1x_1 h_1x_3 h_1x_5

h_3x_{-1} h_3x_1 h_3x_3 h_3x_5

h_1x_{-1} $h_1x_1+h_3x_{-1}$ $h_1x_3+h_3x_1$ $h_3x_3+h_1x_5$ h_3x_5

$w[n'] =$ $\underbrace{h_0x_0+h_1x_{-1}}_{n'=0}$, $\underbrace{h_0x_2+h_2x_0+h_1x_1+h_3x_{-1}}_{n'=1}$, $\underbrace{h_0x_4+h_2x_2+h_1x_3+h_3x_1}_{n'=2}$, $\underbrace{h_2x_4+h_3x_3+h_1x_5}_{n'=3}$

w/o polyphase

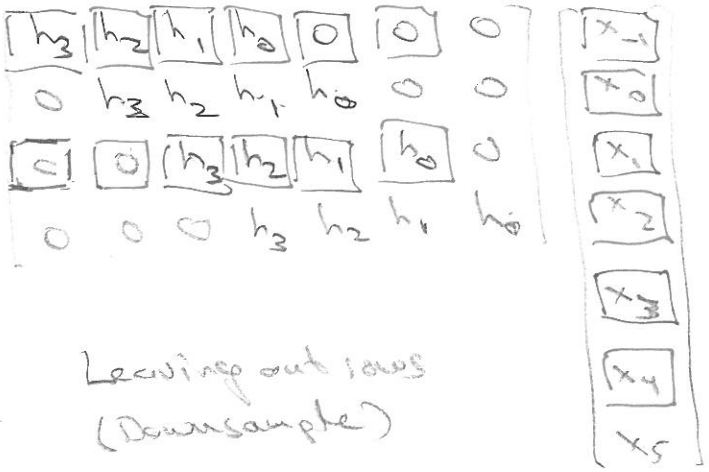
h_0x_{-1} h_0x_0 h_0x_1 h_0x_2 h_0x_3 h_0x_4 h_0x_5

h_1x_{-1} h_1x_0 h_1x_1 h_1x_2 h_1x_3 h_1x_4 h_1x_5

h_2x_{-1} h_2x_0 h_2x_1 h_2x_2 h_2x_3 h_2x_4 h_2x_5

h_3x_{-1} h_3x_0 h_3x_1 h_3x_2 h_3x_3 h_3x_4 h_3x_5

$n'=0$ $n'=1$ $n'=2$ $n'=3$ $n'=4$



$h_3 x_{-1} + h_2 x_0 + h_1 x_1 + h_0 x_2$

$h_3 x_0 + h_2 x_1 + h_1 x_2 + h_0 x_3$

$h_3 x_1 + h_2 x_2 + h_1 x_3 + h_0 x_4$

$h_3 x_2 + h_2 x_3 + h_1 x_4 + h_0 x_5$

$$\begin{bmatrix} h_2 & h_0 & 0 \\ 0 & h_2 & h_0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_2 \\ x_4 \end{bmatrix} + \begin{bmatrix} h_3 & h_1 & 0 \\ 0 & h_3 & h_1 \end{bmatrix} \begin{bmatrix} x_{-1} \\ x_1 \\ x_3 \end{bmatrix}$$

$h_2 x_0 + h_0 x_2 + h_3 x_{-1} + h_1 x_1$

$h_2 x_2 + h_0 x_4 + h_3 x_1 + h_1 x_3$

Filtering and Upsampling by L

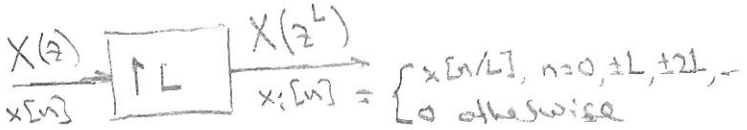
Use $H(z) = \sum_{k=0}^{L-1} E_k(z^L)\, z^{-k}$

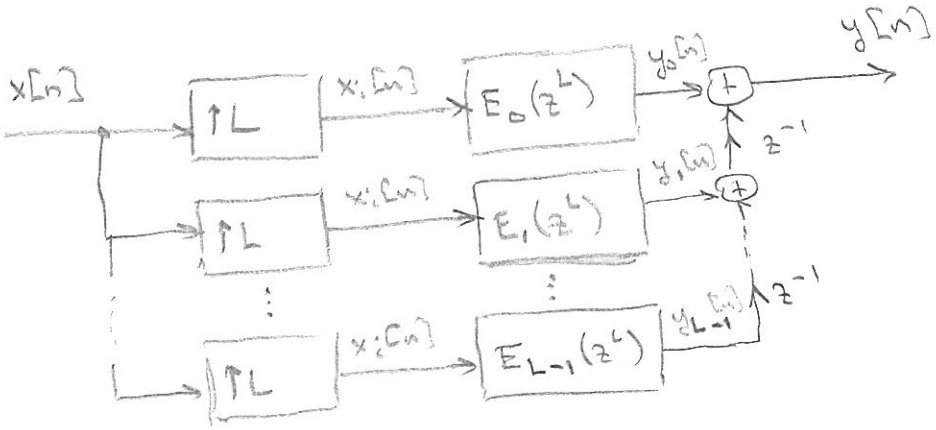
$E_k(z) = \sum_n e_k[n] z^{-n}$

$e_k[n] = h[Ln+k]$

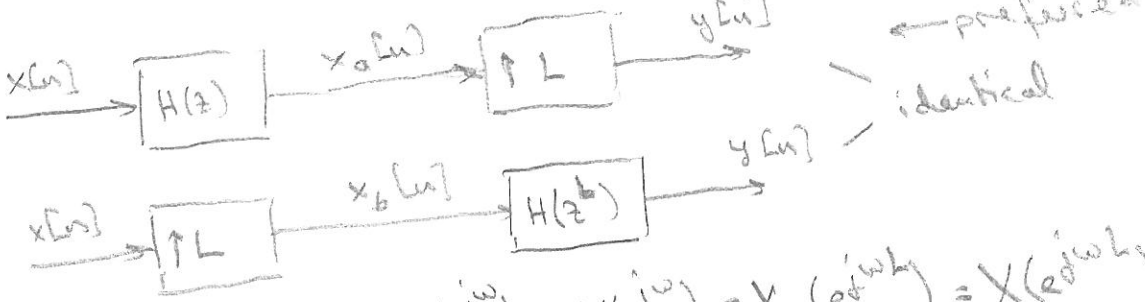
$Y(z) = X(z^L) H(z)$

$= \sum_{k=0}^{L-1} X(z^L) E_k(z^L) z^{-k}$



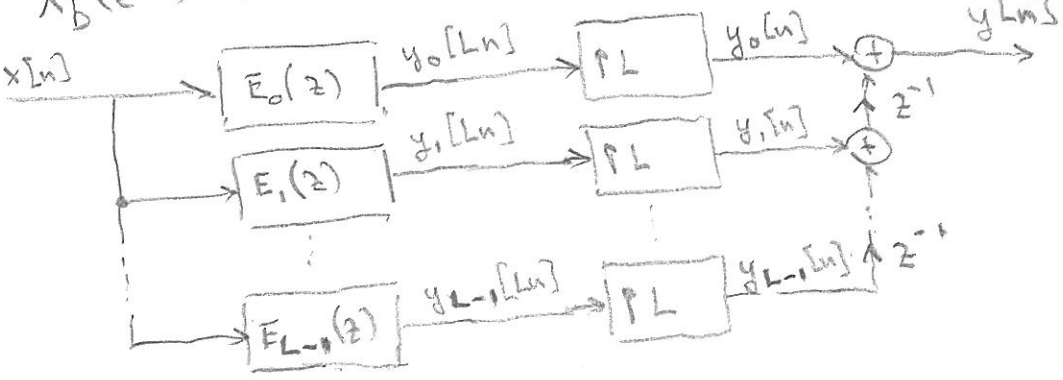


Noble Identities Upsampling

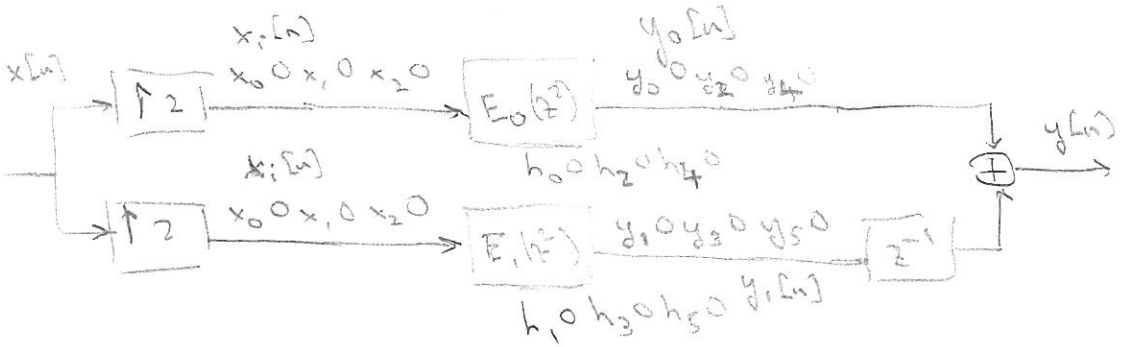


$X_a(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$ $\quad Y(e^{j\omega}) = X_a(e^{j\omega L}) = X(e^{j\omega L}) H(e^{j\omega L})$

$X_b(e^{j\omega}) = X(e^{j\omega L})$ $\quad Y(e^{j\omega}) = X(e^{j\omega L}) H(e^{j\omega L})$



$L = 2$



$h_0x_0 \circ h_0x_1 \circ h_0x_2 \circ$
$\circ \circ \circ \circ \circ \circ$
$h_2x_0 \circ h_2x_1 \circ h_2x_2 \circ$
$\circ \circ \circ \circ \circ \circ$
$h_4x_0 \circ h_4x_1 \circ h_4x_2 \circ$

$h_0x_0 \circ h_0x_1 + h_2x_0 \circ h_0x_2 + h_2x_1 + h_4x_0 \circ h_2x_2 + h_4x_1 \circ h_4x_2$

$h_1x_0 \circ h_1x_1 \circ h_1x_2 \circ$
$\circ \circ \circ \circ \circ \circ$
$h_3x_0 \circ h_3x_1 \circ h_3x_2 \circ$
$\circ \circ \circ \circ \circ \circ$
$h_5x_0 \circ h_5x_1 \circ h_5x_2 \circ$

$h_1x_0 \circ h_1x_1 + h_3x_0 \circ h_1x_2 + h_3x_1 + h_5x_0 \circ h_3x_2 + h_5x_1 \circ h_5x_2$

$h_0x_0, h_1x_0, h_0x_1 + h_2x_0, h_1x_1 + h_3x_0, h_0x_2 + h_2x_1 + h_4x_0, h_1x_2 + h_3x_1 + h_5x_0, h_2x_2 + h_4x_1, h_3x_2 + h_5x_1,$
h_4x_2, h_5x_1

$x_0 \; x_1 \; x_2 \; x_3$ * $h_0 \; h_1 \; h_2 \; h_3 \; h_4 \; h_5$

$h_0x_0 \quad h_0x_1 \quad h_0x_2 \quad h_0x_3$

$h_1x_0 \quad h_1x_1 \quad h_1x_2 \quad h_1x_3$

$h_2x_0 \quad h_2x_1 \quad h_2x_2 \quad h_2x_3$

$h_3x_0 \quad h_3x_1 \quad h_3x_2 \quad h_3x_3$

$h_4x_0 \quad h_4x_1 \quad h_4x_2 \quad h_4x_3$

$h_5x_0 \quad h_5x_1 \quad h_5x_2 \quad h_5x_3$

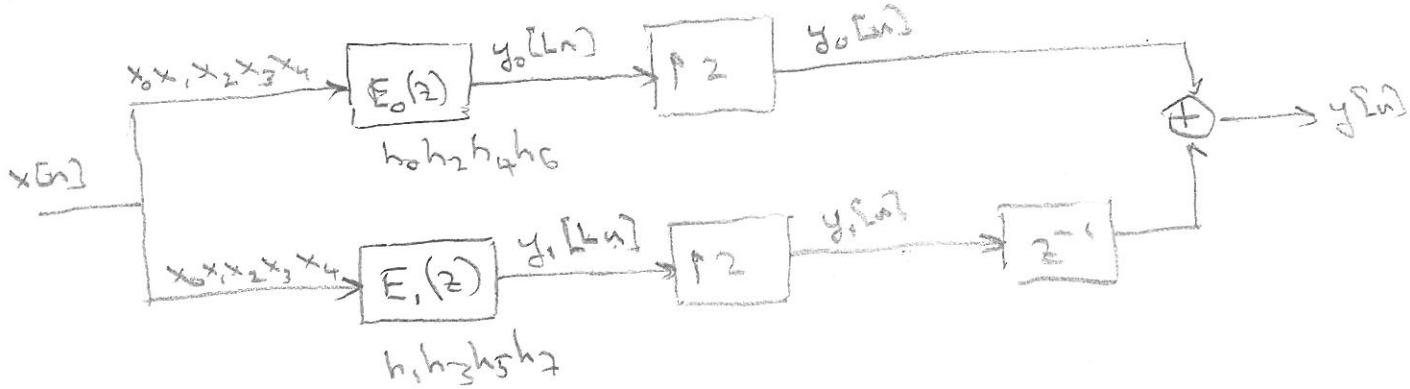
$y[0]$ $y[1]$ $y[2]$

$h_0x_0,\ h_1x_0,\ h_0x_1+h_2x_0,\ h_1x_1+h_3x_0,\ h_0x_2+h_2x_1+h_4x_0,\ h_1x_2+h_3x_1+h_5x_0,\ h_0x_3+h_2x_2+h_4x_1,\ h_1x_3+h_3x_2+h_5x_1$

$y[3]$

$h_2x_3+h_4x_2,\ h_3x_3+h_5x_2,\ h_4x_3,\ h_5x_3$

10-26-20



h_0x_0 h_0x_1 h_0x_2 h_0x_3
h_2x_0 h_2x_1 h_2x_2 h_2x_3
h_4x_0 h_4x_1 h_4x_2 h_4x_3
h_6x_0 h_6x_1 h_6x_2 h_6x_3

$y_0[Ln]$ $h_0x_0, h_0x_1+h_2x_0, h_0x_2+h_2x_1+h_4x_0, h_0x_3+h_2x_2+h_4x_1+h_6x_0, h_2x_3+h_4x_2+h_6x_1, h_4x_3+h_6x_2, h_6x_3$

h_1x_0 h_1x_1 h_1x_2 h_1x_3
h_3x_0 h_3x_1 h_3x_2 h_3x_3
h_5x_0 h_5x_1 h_5x_2 h_5x_3
h_7x_0 h_7x_1 h_7x_2 h_7x_3

$y_1[Ln]$ $h_1x_0, h_1x_1+h_3x_0, h_1x_2+h_3x_1+h_5x_0, h_1x_3+h_3x_2+h_5x_1+h_7x_0, h_3x_3+h_5x_2+h_7x_1, h_5x_3+h_7x_2, h_7x_3$

$y[0] = h_0x_0$ ✓
$y[1] = h_1x_0$ ✓
$y[2] = h_0x_1 + h_2x_0$ ✓
$y[3] = h_1x_1 + h_3x_0$ ✓
$y[4] = h_0x_2 + h_2x_1 + h_4x_0$ ✓
$y[5] = h_1x_2 + h_3x_1 + h_5x_0$ ✓
$y[6] = h_0x_3 + h_2x_2 + h_4x_1 + h_6x_0$
$y[7] = h_1x_3 + h_3x_2 + h_5x_1 + h_7x_0$

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \begin{bmatrix} h_5 & h_4 & h_3 & h_2 & h_1 & h_0 & 0 & 0 & 0 \\ 0 & h_5 & h_4 & h_3 & h_2 & h_1 & h_0 & 0 & 0 \\ 0 & 0 & h_5 & h_4 & h_3 & h_2 & h_1 & h_0 & 0 \\ 0 & 0 & 0 & h_5 & h_4 & h_3 & h_2 & h_1 & h_0 \end{bmatrix} \begin{bmatrix} x_0 \\ 0 \\ x_1 \\ 0 \\ x_2 \\ 0 \\ x_3 \\ 0 \\ x_4 \end{bmatrix}$$

leaving out columns (upsample)

$$y[0] = h_5 x_0 + h_3 x_1 + h_1 x_2$$
$$y[1] = h_4 x_1 + h_2 x_2 + h_0 x_3$$
$$y[2] = h_5 x_1 + h_3 x_2 + h_1 x_3$$
$$y[3] = h_4 x_2 + h_2 x_3 + h_0 x_4$$

$h_5 \; 0 \; h_3 \; 0 \; h_1$ $\qquad$ x_0

$h_5 \; 0 \; h_3 \; 0 \; h_1$

$$\begin{bmatrix} h_5 & h_3 & h_1 & 0 \\ 0 & h_5 & h_3 & h_1 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \begin{bmatrix} h_4 & h_2 & h_0 & 0 \\ 0 & h_4 & h_2 & h_0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$y[0]$ $h_5 x_0 + h_3 x_1 + h_1 x_2$

$y[2]$ $h_5 x_1 + h_3 x_2 + h_1 x_3$

$y[-1]$ $h_4 x_0 + h_2 x_1 + h_0 x_2$

$y[1]$ $h_4 x_1 + h_2 x_2 + h_0 x_3$

$$\begin{bmatrix} h_4 & h_2 & h_0 & 0 \\ 0 & h_4 & h_2 & h_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$y[1]$ $h_4 x_1 + h_2 h_2 + h_0 x_3$

$y[3] = h_4 x_2 + h_2 x_3 + h_0 x_4$